

This Space Peanut Wobbles Like a Top!

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Imagine a space rock shaped like a peanut, **tumbling** through the **asteroid** belt. That's what NASA's Lucy spacecraft found when it zoomed past an **asteroid** called Donaldjohanson last year! Instead of spinning neatly like a top, this **asteroid wobbles** it flips end-over-end once every 10.5 days and sways back and forth every 26.5 days. Scientists say it's like a wobbly toy top that just won't stay still. The **asteroid** is about the size of a small town, and its peanut shape comes from two chunks that gently stuck together after a big crash 155 million years ago. That's young for an **asteroid** many are billions of years old! Lucy's cameras also spotted ridges and **craters**, and even signs that water once flowed there, but only for a short time. The Sun's heat is slowly changing how the **asteroid** spins, a weird effect that can speed up or slow down space rocks. This **flyby** was just a practice run for Lucy's main mission: visiting Jupiter's **Trojan asteroids** in 2027. Those ancient rocks could tell us how the planets formed. For now, Donaldjohanson is giving scientists a rare peek at a young, wobbly world.

Word Treasure

asteroid -- A rocky space object that orbits the Sun, smaller than a planet.

tumbling -- Moving in a clumsy, rolling, or flipping way through space.

craters -- Bowl-shaped dents on a surface, usually made by crashes long ago.

Trojan -- A group of space rocks trapped near a planet's orbit, like Jupiter's.

flyby -- When a spacecraft zooms close to a space object without landing on it.

wobbles -- Shakes or moves unsteadily from side to side instead of spinning smoothly.

Background you need

NASA (the U.S. space agency) built the Lucy spacecraft to explore asteroids, which are rocky leftovers from when the planets formed billions of years ago.

The asteroid belt sits between Mars and Jupiter -- it's like a giant ring of millions of space rocks, some tiny and some as big as small towns.

Donaldjohanson is named after a scientist who studied ancient human bones; Trojan asteroids (space rocks trapped near Jupiter's orbit) might hold clues about Earth's own beginnings.

Break it down

WHO: NASA's Lucy spacecraft and scientists studying space rocks

WHAT: Found that asteroid Donaldjohanson wobbles, has ridges and craters, and once had water flowing on it

WHERE: In the asteroid belt between Mars and Jupiter, on the way to Jupiter's Trojan asteroids

WHY: The flyby was practice for a bigger 2027 mission to study ancient Trojan asteroids and learn how the planets formed

Why it matters

Learning about asteroids helps us understand where Earth came from -- and one day, knowing how they move could help protect our planet from a dangerous crash.

Test what you remember

1. What shape is the asteroid Donaldjohanson?
 - (A) Round like a ball
 - (B) Peanut-shaped
 - (C) Flat like a pancake
 - (D) Cube-shaped
2. How often does the asteroid flip end-over-end?
 - (A) Every 5 days
 - (B) Every 10.5 days
 - (C) Every 26.5 days
 - (D) Every 155 days
3. What did Lucy's cameras spot on the asteroid?
 - (A) Oceans and beaches
 - (B) Ridges and craters
 - (C) Forests and rivers
 - (D) Cities and roads
4. When did the big crash that formed the asteroid happen?
 - (A) 10 years ago
 - (B) 155 million years ago
 - (C) 1 billion years ago
 - (D) In the year 2024
5. What is the Lucy spacecraft's main mission?
 - (A) Landing on the Moon
 - (B) Visiting Jupiter's Trojan asteroids
 - (C) Studying the Sun up close
 - (D) Exploring the surface of Mars
6. What effect does the Sun's heat have on the asteroid?
 - (A) It makes the asteroid glow blue
 - (B) It changes how the asteroid spins
 - (C) It melts the asteroid completely
 - (D) It has no effect at all

Answer key (try the quiz first!): 1: B 2: B 3: B 4: B 5: B 6: B

- 1. Excited space explorers:** Scientists are thrilled because Donaldjohanson is a rare young asteroid -- only 155 million years old -- so it's like getting a baby photo of a space rock. Every wobble and crater teaches them something new.
- 2. Tricky to study:** A wobbly, tumbling asteroid is much harder to map and predict than a steady one. Researchers have to work extra hard to understand what Lucy's pictures really show.
- 3. Practice makes perfect:** This flyby wasn't even the main event -- it was a warm-up. The team wanted to test Lucy's cameras and systems before the bigger mission in 2027, making sure everything works far from Earth.